COMP2711 / DISCRETE MATHEMATICAL TOOLS

Sets and Functions

Past Paper Solutions

68	3	2013-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Updated 30 September 2026

Companion to the existing Sets and Functions practice booklet.

Exercise numbers and question order are unchanged.

READING GUIDE

Contents

Set Operations
12 exercises — S001-S012

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How to use this booklet

Find the exercise number from the companion practice booklet. Questions retain the original section and difficulty order; difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

S001

Set Operations

2024 Fall Midterm | Question 3(b)(i) | Source page 7 | Difficulty 1 | 1 mark
Q-2024F-MID-P03-b-i | Answer provenance: official

> (i) Let $\mathbb{R}$ be the set of real numbers. Provide an example for an infinite set B such that $\mathbb{R} - B$ is countable infinite.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (b) (i) $B = R - N$

Supplementary Explanation

Answer. $B = \mathbb{R} \setminus \mathbb{N}$.

Derivation/proof. Choose $B = \mathbb{R} \setminus \mathbb{N}$. The set B contains every irrational, so it is infinite as required. The difference is $\mathbb{R} \setminus B = \mathbb{R} \setminus (\mathbb{R} \setminus \mathbb{N}) = \mathbb{N}$, which is enumerable and infinite. Thus the construction meets both requirements.

S002

Set Operations

2024 Fall Midterm | Question 3(b)(ii) | Source page 7 | Difficulty 1 | 1 mark
Q-2024F-MID-P03-b-ii | Answer provenance: official

> (ii) Let $\mathbb{R}$ be the set of real numbers. Provide an example for an infinite set B such that $\mathbb{R} - B$ is uncountable.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (ii) $B = (0, 1)$

Supplementary Explanation

Answer. $B = (0, 1)$.

Derivation/proof. Choose $B = (0, 1)$, which contains infinitely many real numbers. Its complement $\mathbb{R} \setminus B = (-\infty, 0] \cup [1, \infty)$ contains $[1, 2]$. Every nondegenerate real interval is uncountable, and a superset of an uncountable set is uncountable. Hence the difference is uncountable. One example suffices; a classification of all possible B is unnecessary.

S003

Set Operations

2022 Midterm | Question 4(a) | Source page 2 | Difficulty 1
Q-2022-MID-P04-a | Answer provenance: official

> Give an example of two uncountable sets A, B such that $A - B$ is:
>
> No Justification is needed.
>
> Finite

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (a) $A = [1, 2), B = (1, 2) \implies A - B = \{1\}$

Supplementary Explanation

Answer. $A = [1, 2), B = (1, 2)$, so $A - B = \{1\}$.

Derivation/proof. Both intervals contain the open interval (1,2), which is uncountable by an affine bijection with (0,1). Their only membership difference is the endpoint 1, included in A and excluded from B; the endpoint 2 is in neither. Hence A−B is exactly the finite singleton {1}.

S004

Set Operations

2022 Midterm | Question 4(c) | Source page 2 | Difficulty 1
Q-2022-MID-P04-c | Answer provenance: official

> Give an example of two uncountable sets A, B such that $A - B$ is:
>
> No Justification is needed.
>
> Uncountable

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (c) $A = [1, 3], B = [1, 2) \implies A - B = [2, 3]$

Supplementary Explanation

Answer. $A = [1, 3], B = [1, 2)$, so $A - B = [2, 3]$.

Derivation/proof. Both A and B contain a nondegenerate real interval, so both are uncountable. Removing [1,2) deletes 1 and all points below2 but leaves2 itself, together with all points through 3. The remaining interval [2,3] contains an affine copy of (0,1), hence is uncountable.

The open interval $(2, 3)$ is in bijection with $(0, 1)$ by translation.

S005

Set Operations

2025 Fall Midterm | Question 3(a) | Source page 10 | Difficulty 2 | 2 marks
Q-2025F-MID-P03-a | Answer provenance: official

> (a) [2 points] Assume $A \subseteq C$ and $B \subseteq C$. Use direct proof to prove that $A \cap B \subseteq C$.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (a) Let $x \in A \cap B$. By definition of intersection, $x \in A$ and $x \in B$. Since $A \subseteq C$, it follows that $x \in C$. Similarly, since $B \subseteq C$, $x \in C$ also holds. Thus, $x \in C$. Therefore, $A \cap B \subseteq C$.

Supplementary Explanation

Answer. $A \cap B \subseteq C$.

Derivation/proof. Take arbitrary $x \in A \cap B$. By the definition of intersection, x belongs to both A and B. Since $A \subseteq C$, from $x \in A$ obtain $x \in C$. Every element of the intersection therefore belongs to C, proving the inclusion. The premise $B \subseteq C$ could be used instead; only one of the two premises is actually needed. An empty intersection satisfies the inclusion vacuously.

S006

Set Operations

2024 Fall Final | Question 2(a) | Source page 5 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P02-a | Answer provenance: official

> (a) Prove or disprove that if $(A \cup B) \subseteq (A \cap B)$, then $A = B$. Note that one approach can be showing that $A \subseteq B$ and $B \subseteq A$, which means that they have to be equal.

Official Answer

Source: physical PDF page 6 (COMP2711_F2024_Final.pdf).

> Answer: (a) To prove $A \subseteq B$, assume $x \in A$. $x \in A \implies x \in A \cup B$ by the definition of union. Then This implies that $x \in A \cap B$ Because $(A \cup B) \subseteq (A \cap B)$. This means $x \in B$ according to the definition of intersection. Therefore, $A \subseteq B$. The same argument can be applied for the other way. So we can prove that $A = B$.

Supplementary Explanation

Answer. True: $A = B$.

Derivation/proof. For any $x \in A$, union membership gives $x \in A \cup B$. The hypothesis forces $x \in A \cap B$, hence $x \in B$. Thus $A \subseteq B$. Conversely, if $x \in B$, then $x \in A \cup B$, so $x \in A \cap B$ and $x \in A$. Therefore $B \subseteq A$. The two-way inclusion proves $A = B$, including cases involving empty sets.

S007

Set Operations

2024 Fall Midterm | Question 3(c) | Source page 7 | Difficulty 2 | 4 marks
Q-2024F-MID-P03-c | Answer provenance: official

> (c) Prove that $A - (B \cap C) = (A - B) \cup (A - C)$. You are Not allowed to use Venn diagram to prove it.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (c) Assume $x \in A - (B \cap C) \leftrightarrow x \in A \wedge x \notin (B \cap C) \leftrightarrow x \in A \wedge (x \notin B \vee x \notin C) \leftrightarrow (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \leftrightarrow (x \in (A - B)) \vee (x \in (A - C)) \leftrightarrow x \in ((A - B) \cup (A - C))$.

Supplementary Explanation

Answer. $A - (B \cap C) = (A - B) \cup (A - C)$.

Derivation/proof. For arbitrary x, membership $x \in A \setminus (B \cap C)$ is equivalent to $x \in A \wedge \neg(x \in B \wedge x \in C)$. De Morgan gives $x \in A \wedge (x \notin B \vee x \notin C)$. The distributive law turns this into $(x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)$, exactly $x \in (A \setminus B) \cup (A \setminus C)$. Every step is a two-way equivalence and x was arbitrary, so the sets have exactly the same members without a Venn diagram.

S008

Set Operations

2023 Fall Midterm | Question 3(a) | Source page 7 | Difficulty 2 | 4 marks
Q-2023F-MID-P03-a | Answer provenance: official

> Let S and T denote two sets. Prove the following by contradiction:
>
> If $S \cap T = \varnothing$ and $S \cup T = T$ then $S = \varnothing$.

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Midterm.pdf).

> (a) Assume that S is not an empty set. This implies that $\exists x$ s.t. $x \in S$. If $x \in S$, then $x \in S \cup T$ or $x \in T$ (because $S \cup T = T$). However, $x \in S \cap T$ contradicts with the fact that $S \cap T = \varnothing$. Consequently, we must have $S = \varnothing$.

Supplementary Explanation

Answer. $S = \varnothing$.

Derivation/proof. Assume $S \cap T = \varnothing$ and $S \cup T = T$, but suppose $S \neq \varnothing$. Then there is an element $x \in S$. It follows that $x \in S \cup T$, and the union equality implies $x \in T$. Hence $x \in S \cap T$, contradicting the premise that this intersection is empty. The assumption $S \neq \varnothing$ must therefore be false, proving the conclusion by contradiction.

S009

Set Operations

2022 Midterm | Question 4(b) | Source page 2 | Difficulty 2
Q-2022-MID-P04-b | Answer provenance: official

> Give an example of two uncountable sets A, B such that $A - B$ is:
>
> No Justification is needed.
>
> Countable infinite

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (b) $A = (-\infty, -1) \cup \mathbb{N}, B = (-\infty, -1) \implies A - B = \mathbb{N}$

Supplementary Explanation

Answer. $A = (-\infty, -1) \cup \mathbb{N}, B = (-\infty, -1)$, so $A - B = \mathbb{N}$.

Derivation/proof. The negative interval B is uncountable, and A contains it so is also uncountable. Natural numbers do not lie below −1, so the two parts of A are disjoint. Removing B therefore leaves exactly the naturals, a countably infinite set. This works whether zero belongs to the convention for naturals or not.

The remainder is countable and infinite, regardless of whether the convention includes zero in $\mathbb{N}$.

S010

Set Operations

2021 Midterm | Question 4(b) | Source page 2 | Difficulty 2 | 2 marks
Q-2021-MID-P04-b | Answer provenance: official

> Let $\mathbb{N}$ be the set of all natural numbers.
>
> Is $\{S \mid S \subseteq \mathbb{N}, |S| \text{ is infinite}\}$ countable or uncountable? Justification is not necessary.

Official Answer

Source: physical PDF page 2 (2021-midterm.pdf).

> (b) Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. The full power set is uncountable by Cantor's diagonal argument. Its finite-subset subfamily is countable by enumeration using the maximum element. If its infinite-subset subfamily were also countable, the union of these two countable families would be countable, contradicting uncountability of the power set. Thus the infinite-subset family is uncountable.

Writing the two families as F and I gives $P(\mathbb{N}) = F \cup I$. If both families were countable, interleaving their enumerations would enumerate the power set. A family is not uncountable merely because each member is infinite.

If both families were countable, interleaving the enumerations would enumerate their union. Hence I is uncountable.

S011

Set Operations

2020 Final Part 2 | Question 2 | Source page 2 | Difficulty 2 | 5 marks
Q-2020-FINAL2-P02 | Answer provenance: official

> Three seminars were held by CSE department last week. 70% of students attended the seminar on Monday, 80% of students attended the seminar on Wednesday, and 90% of students attended the seminar on Friday. Prove that at least 40% of students attended all three seminars.

Official Answer

Source: physical PDF page 2 (2020-final-part-2.pdf).

U - Set of all students; A - Set of students having attended the seminar on Monday, $|A| = 70\%|U|$; B - Set of students having attended the seminar on Wednesday, $|B| = 80\%|U|$; C - Set of students having attended the seminar on Friday, $|C| = 90\%|U|$.

$$\begin{aligned} |A \cap B \cap C| &= |U| - |\overline{A} \cup \overline{B} \cup \overline{C}| \\ &\geq |U| - (|\overline{A}| + |\overline{B}| + |\overline{C}|) \\ &= |U| - (30\%|U| + 20\%|U| + 10\%|U|) \\ &= 40\%|U|. \end{aligned}$$

The inequality uses the fact that $|X \cup Y \cup Z| \leq |X| + |Y| + |Z|$.

Supplementary Explanation

Answer. At least 40% attended all three.

Derivation/proof. Let U be the students and A,B,C the three attendance sets. Their complements have sizes .3|U|,.2|U|,.1|U|. Anyone not attending all three lies in $\overline{A} \cup \overline{B} \cup \overline{C}$, whose size is at most the sum .6|U|. Therefore $|A \cap B \cap C| = |U| - |\overline{A} \cup \overline{B} \cup \overline{C}| \geq .4|U|$. No independence between attendance events is needed.

The union of the three absence sets has size at most $(0.3 + 0.2 + 0.1)|U| = 0.6|U|$. Taking its complement yields $|A \cap B \cap C| \geq |U| - 0.6|U| = 0.4|U|$, so at least forty percent attend all three.

S012

Set Operations

2020 Final Part 2 | Question 3 | Source page 2 | Difficulty 2 | 5 marks
Q-2020-FINAL2-P03 | Answer provenance: official

> Let A and B be two events. It is known that $p(A) = 0.7$, $p(B) = 0.6$, $p(A \cap B) = 0.5$, what is $p(B \mid A \cup \overline{B})$?

Official Answer

Source: physical PDF page 2 (2020-final-part-2.pdf).

$$p(B \mid A \cup \overline{B}) = \frac{p(B \cap (A \cup \overline{B}))}{p(A \cup \overline{B})}.$$

Since $B \cap (A \cup \overline{B}) = (B \cap A) \cup (B \cap \overline{B}) = (B \cap A) \cup \varnothing = B \cap A$,

$$p(B \cap (A \cup \overline{B})) = p(B \cap A) = 0.5.$$

$$p(A \cup \overline{B}) = 1 - p(\overline{A} \cap B) = 1 - (p(B) - p(A \cap B)) = 0.9.$$

The final result is,

$$p(B \mid A \cup \overline{B}) = \frac{p(B \cap (A \cup \overline{B}))}{p(A \cup \overline{B})} = \frac{5}{9}.$$

Supplementary Explanation

Answer. $5/9$.

Derivation/proof. The conditional numerator is $p(B \cap (A \cup \overline{B})) = p(A \cap B) = .5$, since $B \cap \overline{B} = \varnothing$. The conditioning event excludes exactly $\overline{A} \cap B$, of probability $.6 - .5 = .1$. Its probability is therefore $.9$. Dividing gives $.5/.9 = 5/9$. The bar over B is visible in the printed prompt and materially changes both event calculations.

The probability of the excluded event is $0.6 - 0.5 = 0.1$, leaving conditioning probability 0.9. The conditional numerator is 0.5, hence the quotient is $0.5/0.9 = 5/9$.

S013

Functions

2020 Midterm | Question 5(e) | Source page 3 | Difficulty 1
Q-2020-MID-P05-e | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> The set of all functions from $\{0, 1\}$ to $\mathbb{N}$

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

Countable.

Supplementary Explanation

Answer. Countable.

Derivation/proof. Each function corresponds bijectively to the ordered pair $(f(0), f(1)) \in \mathbb{N}^2$. Pairs can be enumerated by increasing coordinate sum, with finitely many pairs at each sum. Thus this function space is countable; unlike an infinite-domain function, it contains only two independent output choices per object.

Diagonal enumeration is one available proof of countability, not a required unique method.

The map $f \mapsto (f(0), f(1))$ is a bijection. Ordering the pairs first by $f(0) + f(1)$ and then within each finite sum gives a valid diagonal enumeration, proving countability.

S014

Functions

2018 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2018-MID-P04-a-contrapositive | Answer provenance: official

> Consider finite sets S and T, and a function $f : S \to T$.
>
> "If f is injective, then $|T| \geq |S|$."
>
> (a) Consider finite sets S and T, and a function $f : S \to T$. Write down the contrapositive of "If f is injective, then $|T| \geq |S|$."

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

The contrapositive is "If $|T| < |S|$, then f is not injective". We can prove it by the pigeonhole principle. f maps pigeons in S to holes in T. Since $|T| < |S|$, there must be two pigeons that share a same hole, say $a, b \in S$. Then $a \neq b$ but $f(a) = f(b)$, so f is not injective.

Supplementary Explanation

Answer. If $|T| < |S|$, then f is not injective.

Derivation/proof. Write the original statement as $A \to B$, where A is injectivity and B is $|T| \geq |S|$. Its contrapositive is $\neg B \to \neg A$. For finite cardinalities, $\neg(|T| \geq |S|)$ is exactly $|T| < |S|$, yielding the stated sentence.

S015

Functions

2013 Midterm | Question 2(a) | Source page 4 | Difficulty 1
Q-2013-MID-P02-a | Answer provenance: official

> Consider $S_3 = \{1, 2, 3\}$ and $S_n = \{1, 2, \ldots, n\}$, where $n > 3$.
>
> (a) How many different functions f are there from S_3 to S_n?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2013-midterm.pdf).

(a) To define f are from S_3 to S_n, we need to specify $f(1)$, $f(2)$, and $f(3)$. Each of those values can be any of the n elements in S_n. So, the answer is: n^3.

Supplementary Explanation

Answer. n^3.

Derivation/proof. A function is fixed by the independent choices of $f(1)$, $f(2)$, $f(3)$. Each has n possible images, hence $n \cdot n \cdot n = n^3$.

Repeated images are permitted. Each ordered triple of images determines exactly one function.

S016

Functions

2013 Midterm | Question 2(b) | Source page 4 | Difficulty 1
Q-2013-MID-P02-b | Answer provenance: official

> Consider $S_3 = \{1, 2, 3\}$ and $S_n = \{1, 2, \ldots, n\}$, where $n > 3$.
>
> (b) How many different functions f are there from S_3 to S_n such that $f(1) \neq f(2)$?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2013-midterm.pdf).

(b) We determine $f(1)$, $f(2)$ and $f(3)$ in order. There are n possibilities for $f(1)$, $n - 1$ possibilities for $f(2)$, and n possibilities for $f(3)$. So, the answer is: $n^2(n - 1)$.

Supplementary Explanation

Answer. $n^2(n - 1)$.

Derivation/proof. Choose $f(1)$ in n ways, $f(2)$ in $n - 1$ ways different from it, and $f(3)$ freely in n ways.

The third image may equal either previous image, so the product is $n(n - 1)n = n^2(n - 1)$.

There are n−1 choices for the second image after the first is fixed.

S017

Functions

2013 Midterm | Question 2(c) | Source page 4 | Difficulty 1
Q-2013-MID-P02-c | Answer provenance: official

> Consider $S_3 = \{1, 2, 3\}$ and $S_n = \{1, 2, \ldots, n\}$, where $n > 3$.
>
> (c) How many different functions f are there from S_3 to S_n such that $f(1) \neq f(2)$ and $f(2) \neq f(3)$?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2013-midterm.pdf).

(c) We determine $f(2)$, $f(1)$ and $f(3)$ in order. There are n possibilities for $f(2)$, $n-1$ possibilities for $f(1)$, and $n-1$ possibilities for $f(3)$. So, the answer is: $n(n-1)^2$.

Supplementary Explanation

Answer. $n(n-1)^2$.

Derivation/proof. Choose the middle value $f(2)$ first (n choices). Each endpoint must avoid it, independently giving $(n-1)^2$ choices; $f(1)$ and $f(3)$ may coincide.

S018

Functions

2020 Midterm | Question 5(d) | Source page 3 | Difficulty 2
Q-2020-MID-P05-d | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> The set of all functions from $\mathbb{N}$ to $\mathbb{N}$

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Choose two distinct naturals a,b. For every subset S define $f_S(i) = a$ when $i \in S$, and b otherwise. Distinct subsets differ at some i and thus yield distinct functions. This injects the uncountable power set of the naturals into the function space, proving that the full function space is uncountable independently of whether zero is considered natural.

Any two distinct natural numbers suffice for the encoding, so no convention about whether zero is natural is needed.

Choose distinct naturals a, b and define $f_S(i) = a$ for membership in S and $f_S(i) = b$ otherwise. If $S \neq T$, a coordinate in their symmetric difference yields $f_S \neq f_T$. This injective embedding transfers uncountability to the function space.

S019

Functions

2018 Midterm | Question 4(a) | Source page 6 | Difficulty 2
Q-2018-MID-P04-a-proof | Answer provenance: official

> Consider finite sets S and T, and a function $f : S \to T$.
>
> "If f is injective, then $|T| \geq |S|$."
>
> Then, prove the contrapositive directly.

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

The contrapositive is "If $|T| < |S|$, then f is not injective". We can prove it by the pigeonhole principle. f maps pigeons in S to holes in T. Since $|T| < |S|$, there must be two pigeons that share a same hole, say $a, b \in S$. Then $a \neq b$ but $f(a) = f(b)$, so f is not injective.

Supplementary Explanation

Answer. If $|T| < |S|$, the function f is not injective.

Derivation/proof. Assume $|S| > |T|$. Regard the elements of S as pigeons and the elements of T as holes, sending s to hole $f(s)$. By pigeonhole principle, two distinct elements $a, b \in S$ share an image: $f(a) = f(b)$. This is precisely failure of injectivity.

To show failure of injectivity, the two inputs must satisfy $a \neq b$ while their outputs agree; equality of outputs for the same input would prove nothing.

S020

Functions

2018 Midterm | Question 4(b) | Source page 6 | Difficulty 2
Q-2018-MID-P04-b | Answer provenance: official

> (b) For functions on finite sets $f : S \to T, g : T \to U$, if f is injective, and g is surjective, is it true that $g \circ f$ must be injective? Prove it.

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

False. Students can provide any counterexample such that f is injective, g is surjective, and $g \circ f$ is not injective. For example: $S = T = \{a, b\}, U = \{0\}, f(x) = x, g(y) = 0$, then $g(f(x)) = 0$. It is not injective because $g(f(a)) = g(f(b)) = 0$.

Supplementary Explanation

Answer. No. Let $S = T = \{a, b\}, U = \{0\}$, f be the identity and g be constant zero.

Derivation/proof. The identity sends distinct inputs to distinct elements, so f is injective. The image of g is all of the singleton U, so g is surjective. Yet $g(f(a)) = 0 = g(f(b))$ with $a \neq b$, hence the composite is not injective. Every required premise is satisfied by this counterexample.

S021

Functions

2013 Midterm | Question 2(d) | Source page 4 | Difficulty 2
Q-2013-MID-P02-d | Answer provenance: official

> Consider $S_3 = \{1, 2, 3\}$ and $S_n = \{1, 2, \ldots, n\}$, where $n > 3$.
>
> (d) How many different functions f are there from S_3 to S_n such that $f(1) \neq f(2)$, $f(2) \neq f(3)$, and $f(1) < f(3)$?
>
> Explain how your answers are derived.

Official Answer

Source: PDF physical page 4 (2013-midterm.pdf).

(d) We determine $f(2)$, $f(1)$ and $f(3)$ in order. There are n possibilities for $f(2)$. After $f(2)$ is determined, there are $n-1$ numbers remaining. We need to choose two of them as values for $f(1)$ and $f(3)$. There are $\binom{n-1}{2}$ different ways to choose the two numbers. So, the answer is: $n\binom{n-1}{2}$.

Supplementary Explanation

Answer. $n\binom{n-1}{2}$.

Derivation/proof. Pick $f(2)$ first. From the remaining $n-1$ values choose the two distinct endpoint values; their increasing order uniquely assigns the smaller to $f(1)$ and the larger to $f(3)$.

The increasing endpoint assignment is unique, so there is no factor of two.

S022

Functions

2025 Spring Final | Question 1(a)(iii) | Source page 2 | Difficulty 3 | 1 mark
Q-2025S-FINAL-P01-a-iii | Answer provenance: official

> (a) [3 pts, 1 each] Determine whether the following statements are True or False. No argument is needed. $\mathcal{P}(X)$ denotes the power set of X.
>
> (iii) There is a bijection between all the real numbers of $[0, 1]$ and all the real numbers of $[0, 1)$.

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(a) iii) T

Supplementary Explanation

Answer. True.

Derivation/proof. Define $f(1/n) = 1/(n+1)$ for $n = 1, 2, \ldots$, and set $f(x) = x$ for all other $x \in [0, 1]$. In particular, 1 goes to $1/2$ and 0 stays 0, so the image is contained in $[0, 1)$. The shifted reciprocal values are distinct, while unchanged points lie outside that sequence, so no images collide and f is injective. For a target $y = 1/m$ with $m \geq 2$, the preimage is $1/(m-1)$; every other target is its own preimage. Thus f is surjective and is the required bijection.

S023

Functions

2024 Fall Midterm | Question 7 | Source page 12 | Difficulty 3 | 5 marks
Q-2024F-MID-P07 | Answer provenance: official

> Assume that there are 5 processing jobs that must be assigned to 3 servers. Use Inclusion-Exclusion to calculate how many assignments exist such that each server is assigned to at least one job? Briefly explain why.

Official Answer

Source: physical PDF page 12 (COMP2711_F2024_Midterm.pdf).

> Answer: Solution 1: This is the number of onto functions from a set of 5 elements to a set of 3 elements: $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5$ Solution 2:

Supplementary Explanation

Answer. $3^5 - \binom{3}{1}2^5 + \binom{3}{2}1^5 = 150$.

Derivation/proof. Each of five jobs has three choices, giving $3^5 = 243$ functions. Let A_i consist of assignments leaving server i empty; then $|A_i| = 2^5$. If two specified servers are empty, all jobs go to the remaining server, so the intersection has size $1^5 = 1$. The triple intersection is empty because five jobs cannot use no server. Inclusion-exclusion gives $243 - 3 \cdot 32 + 3 \cdot 1 = 150$. Jobs and servers have identities, so these are labelled assignments, not unordered partitions.

S024

Functions

2018 Midterm | Question 4(c) | Source page 6 | Difficulty 3
Q-2018-MID-P04-c | Answer provenance: official

> (c) Consider a lossless data compression algorithm, like zip or rar: compressed files can always be decompressed to obtain the original files. Prove that it is impossible for a lossless data compression algorithm to reduce the size of every file. (Hint: Use (a), and prove by contradiction on the set of all files of size n bits or less.)

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

By contradiction, consider such a function f defined on $F_n \to T$, where F_n is the set of all files of size n bits or less. As f reduces the sizes of all files, T satisfies $T \subseteq F_{n-1}$, the size of all files of size $n-1$ bits or less. Clearly $F_{n-1} \subset F_n$, so $|T| < |F_n|$. However, since f is invertible, f must be injective. Therefore, using (a), we have $|T| \geq |F_{n-1}|$. This is a contradiction.

Supplementary Explanation

Answer. No lossless compressor can shorten every file.

Derivation/proof. Let F_n contain all bit strings of length at most n, including the empty string. Then $|F_n| = \sum_{i=0}^{n} 2^i = 2^{n+1} - 1$, whereas $|F_{n-1}| = 2^n - 1$. If every input became shorter, compression would map F_n into F_{n-1}. Lossless decoding implies injectivity: equal compressed outputs must decode to equal inputs. This contradicts pigeonhole principle because $|F_n| > |F_{n-1}|$.

More explicitly, $|T| \leq 2^n - 1 < 2^{n+1} - 1 = |F_n|$, while lossless decoding requires $|T| \geq |F_n|$.

A decoder must reconstruct every original string from its compressed representation. Thus two distinct inputs cannot share one output; injectivity and the pigeonhole principle supply the cardinality obstruction.

S025

Functions

2017 Final | Question 1 | Source page 3 | Difficulty 3 | 4 marks
Q-2017-FINAL-P01 | Answer provenance: official

> An office manager has four employees and nine reports to be done. In how many ways can the reports be assigned to the employees so that each employee has at least one report to do?

Official Answer

Source: physical PDF page 3 (2017-final.pdf).

$4^9 - \binom{4}{1}3^9 + \binom{4}{2}2^9 - \binom{4}{3}1^9$.

Supplementary Explanation

Answer. $4^9 - \binom{4}{1}3^9 + \binom{4}{2}2^9 - \binom{4}{3}1^9 = 186480$.

Derivation/proof. There are 4^9 unrestricted assignments. For each employee, define the bad event that they receive no report. If a specified set of j employees is omitted, each report has $4 - j$ choices, giving $(4 - j)^9$ assignments. Inclusion-exclusion gives $\sum_{j=0}^{4}(-1)^j\binom{4}{j}(4 - j)^9$, whose last term is zero. Evaluating yields 186480.

Evaluation gives $262144 - 78732 + 3072 - 4 = 186480$. Each nonempty assignment survives inclusion-exclusion exactly once.

S026

Functions

2021 Midterm | Question Bonus | Source page 4 | Difficulty 4 | 10 marks
Q-2021-MID-BONUS | Answer provenance: derived

> Prove $|(0,1)| = |(0,1) \times (0,1)|$.

Official Answer

Source: physical PDF page 4 (2021-midterm.pdf), physical PDF page 5 (2021-midterm.pdf).

> We need to construct an injection from $(0,1)$ to $(0,1) \times (0,1)$, as well as one the other way round. Then applying the Schröder-Bernstein theorem would complete the proof. An injection $f : (0,1) \to (0,1) \times (0,1)$ is trivial, e.g., $f(x) = (x, 0)$. An injection $f : (0,1) \times (0,1) \to (0,1)$ can be defined as follows. For any $x, y \in (0,1)$, let $x = 0.x_1x_2\cdots$ and $y = 0.y_1y_2\cdots$ be their valid decimal representations (i.e., not ending with infinitely many 9's). Define $f(x,y) = 0.x_1y_1x_2y_2\cdots$. Note that this must also be a valid decimal representation of some real number in $(0,1)$, because if it's not, $x = 0.x_1x_2\cdots$ and $y = 0.y_1y_2\cdots$ must end with infinitely many 9's. For any $(x,y) \neq (x',y')$, they must have at least one different digit, so $f(x,y) \neq f(x',y')$, hence f is an injection.

Supplementary Explanation

Answer. The interval and its Cartesian square have equal cardinality.

Derivation/proof. An injection into the square is f(x)=(x,1/2). For the reverse direction choose the unique decimal representations of x,y not eventually all 9s, and interleave their digits: g(x,y)=0.x1y1x2y2…. This number is positive because at least one input digit is positive, and less than one because the interleaving is not all 9s. It is not eventually all 9s, since that would force both input expansions eventually all 9s. Thus its canonical digits uniquely recover both inputs, proving injectivity. Cantor–Schröder–Bernstein now gives a bijection and equality of cardinalities.

Equality of images under $f(x) = (x, 1/2)$ forces equality of first coordinates. For the reverse map, odd-position digits recover x and even-position digits recover y.

For $g(x,y) = 0.x_1y_1x_2y_2\cdots$, odd positions recover x and even positions recover y. If its digits were eventually all nines, both input expansions would also be eventually all nines, contrary to the canonical choice.

S027

Countability

2022 Midterm | Question 5(d) | Source page 2 | Difficulty 1
Q-2022-MID-P05-d | Answer provenance: official

> For each of the following sets, determine if it is countable or uncountable. No justification is needed.
>
> The set of circles in the plane.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (d) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Fix the center at the origin. For each real $r>0$ take the circle $x^2+y^2=r^2$. Distinct radii yield distinct circles, so $r \mapsto$ circle is an injection from the uncountable positive real interval into the set of all circles. A set containing this uncountable subfamily must itself be uncountable; no enumeration of all centers is needed.

The equation is $x^2 + y^2 = r^2$. Distinct radii give distinct circles: the point $(r, 0)$ has distance exactly r from the origin. The positive real numbers therefore inject into the circle family.

S028

Countability

2020 Midterm | Question 5(a) | Source page 3 | Difficulty 1
Q-2020-MID-P05-a | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> $\mathbb{N}^3$

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

Countable.

Supplementary Explanation

Answer. Countable.

Derivation/proof. Group triples by their coordinate sum. Each fixed sum contains only finitely many natural-number triples, and every triple belongs to one such group. Listing these finite groups in increasing sum enumerates the entire product. It is infinite because fixing two coordinates leaves infinitely many choices for the third.

The set is infinite by fixing two coordinates and varying the first. The argument works whether natural numbers start at zero or one.

Group the triples by $a + b + c = s$. There are finitely many triples for each fixed s, so listing these finite groups in increasing s enumerates the product. Fixing two coordinates and varying the third proves that the set is countably infinite rather than finite.

S029

Countability

2019 Midterm | Question 7(a) | Source page 9 | Difficulty 1 | 2 marks
Q-2019-MID-P07-a | Answer provenance: official

> Give an example of two uncountable sets A and B such that $A - B$ is
>
> (a) finite.

Official Answer

Source: physical PDF page 9 (2019-midterm.pdf).

In each case, let us take A to be the set of real numbers. We can let B be the set of real numbers as well; then $A - B = \varnothing$, which is finite.

Supplementary Explanation

Answer. $A = B = \mathbb{R}$, so $A - B = \varnothing$.

Derivation/proof. The real numbers are uncountable, so both chosen inputs satisfy the requirement. Every element of A also belongs to B, leaving no element in the difference. The empty set has cardinality zero, which is finite. The problem does not require A and B to be distinct.

Neither distinct inputs nor a nonempty finite difference are required.

S030

Countability

2025 Fall Final | Question 2(a)(i) | Source page 5 | Difficulty 2 | 1 mark
Q-2025F-FINAL-P02-a-i | Answer provenance: official

> (a) [3 points, 1 pt each] Determine whether each of these sets is finite, countably infinite, or uncountable. No argument is needed.
>
> (i) The set of binary strings of infinite length with no two identical consecutive bits (i.e., no "00" and no "11" anywhere in the sequence).

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Final.pdf).

(a) (i) Countable infinite and Finite are both accepted here. based on the definition, the correct answer is finite, because there are only 2 such strings. However if the phrasing of the question was set of arbitrary length strings ... then the answer was countable infinite. As it was not discussed clearly, we accept both Countable Infinite and Finite answers.

Supplementary Explanation

Answer. Finite; exactly two strings.

Derivation/proof. If the first bit is zero, avoiding "00" forces the next bit to be one. Avoiding "11" then forces the following bit to be zero. Continuing gives the unique string 010101 . . .. Starting with one similarly forces 101010 . . .. There are only two first-bit choices, and both distinct strings satisfy the restrictions; the cardinality is exactly two. The marking note also accepts "countably infinite" for an alternative interpretation involving arbitrary finite lengths. The literal question specifies infinite length, for which the mathematical classification is finite.

S031

Countability

2025 Fall Final | Question 2(a)(ii) | Source page 5 | Difficulty 2 | 1 mark
Q-2025F-FINAL-P02-a-ii | Answer provenance: official

> (a) [3 points, 1 pt each] Determine whether each of these sets is finite, countably infinite, or uncountable. No argument is needed.
>
> (ii) The set of ternary strings (i.e., they are made of $\{0, 1, 2\}$) of infinite length.

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Final.pdf).

(ii) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Suppose all infinite ternary strings can be listed as $s_1, s_2, \ldots$. Define the nth digit of a new string by $d_n = (s_n(n) + 1) \bmod 3$. Every digit remains in $\{0, 1, 2\}$, while $d_n \neq s_n(n)$. The new string therefore differs from the nth listed string at position n, for every n. It is absent from the proposed complete list, a contradiction. Hence the set is uncountable.

S032

Countability

2025 Fall Final | Question 2(a)(iii) | Source page 5 | Difficulty 2 | 1 mark
Q-2025F-FINAL-P02-a-iii | Answer provenance: official

> (a) [3 points, 1 pt each] Determine whether each of these sets is finite, countably infinite, or uncountable. No argument is needed.
>
> (iii) The set of ternary strings of infinite length such that they contain "012".

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Final.pdf).

(iii) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. For every infinite ternary string s, define $f(s) = 012s$ by attaching the fixed prefix 012. This new string contains the required consecutive substring. If $s \neq t$, their tails differ, so $012s \neq 012t$; thus f is injective. The set of infinite ternary strings is uncountable, so its injective image is uncountable. That image is contained in the requested set, proving that the requested set is also uncountable. Requiring an occurrence of 012 leaves the infinite tail unrestricted.

S033

Countability

2025 Fall Midterm | Question 3(c)(i) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-i | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (i) for any set $S \subset \mathbb{N}$, $\mathcal{P}(S)$ is uncountable infinite.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (i) F

Supplementary Explanation

Answer. False.

Derivation/proof. Take $S = \varnothing \subset \mathbb{N}$. Its power set $\mathcal{P}(S) = \{\varnothing\}$ has only one element and is finite, so it is not uncountably infinite. One valid counterexample disproves the universal claim. Even if S must be nonempty, a singleton has a two-element power set.

S034

Countability

2025 Fall Midterm | Question 3(c)(ii) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-ii | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (ii) The set of functions from $\mathbb{N} \to \mathbb{N}$ is uncountable infinite.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (ii) T

Supplementary Explanation

Answer. True.

Derivation/proof. Suppose all functions can be listed as $f_0, f_1, \ldots$. Define $g(n) = f_n(n) + 1$. Successor closure ensures g is still a function from natural numbers to natural numbers, but $g(n) \neq f_n(n)$ for each n, so $g \neq f_n$. Thus g is absent from the supposedly complete list, a contradiction.

S035

Countability

2025 Fall Midterm | Question 3(c)(iii) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-iii | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (iii) Assume A and B are both uncountable sets such that $A \subset B$. $B - A$ is either countable infinite or finite.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (iii) F

Supplementary Explanation

Answer. False.

Derivation/proof. Take $A = (-\infty, 0)$ and $B = \mathbb{R}$. Both sets are uncountable and A is a proper subset of B, yet $B - A = [0, \infty)$ contains (0,1) and is uncountable. Thus the difference is neither finite nor countably infinite. Subtraction of sets does not perform numerical subtraction of their cardinalities.

S036

Countability

2025 Fall Midterm | Question 3(c)(iv) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-iv | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (iv) Set S is countable infinite if there exists a bijection between S and the set of integers $\mathbb{Z}$.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (iv) T

Supplementary Explanation

Answer. True.

Derivation/proof. Enumerate the integers as $0, 1, -1, 2, -2, \ldots$. Given a bijection $f : S \to \mathbb{Z}$, list the unique preimage of each integer in this order: $f^{-1}(0), f^{-1}(1), f^{-1}(-1), \ldots$. Injectivity and surjectivity ensure exactly one preimage per integer and that every element of S appears, without repetition. Distinct infinitely many integers give distinct preimages, so S is countably infinite.

S037

Countability

2025 Fall Midterm | Question 3(c)(v) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-v | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (v) The following set is countable infinite $\{0, 00, 000, 0000, \cdots\}$.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (v) T

Supplementary Explanation

Answer. True.

Derivation/proof. Map a string of n zeros to its positive integer length: $0^n \mapsto n$, for $n \geq 1$. Different lengths give different strings, and each positive length determines exactly one all-zero string. This is a bijection with the positive integers, so the set is countably infinite. The strings 0 and 00 must not be identified as the same numerical zero; each string is finite but lengths are unbounded.

S038

Countability

2025 Fall Midterm | Question 3(c)(vi) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-vi | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (vi) The set of binary strings of infinite length is uncountable.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (vi) T

Supplementary Explanation

Answer. True.

Derivation/proof. If the infinite strings were listed as $s_1, s_2, \ldots$, define the nth bit of a new string by $d_n = 1 - s_n(n)$. Each bit is still zero or one, so this is a valid infinite binary string. It differs at position n from s_n, for every n, and therefore is missing from the list. This contradicts complete enumeration and proves uncountability.

S039

Countability

2025 Fall Midterm | Question 3(c)(vii) | Source page 11 | Difficulty 2 | 1 mark
Q-2025F-MID-P03-c-vii | Answer provenance: official

> (c) [7 points, 1 pt each] Determine whether the following statements are True or False. No argument is needed.
>
> (vii) The set of binary strings of infinite length such that it has exactly three “1”'s is countable infinite.

Official Answer

Source: physical PDF page 11 (COMP2711_F2025_Midterm.pdf).

> (c) (vii) T

Supplementary Explanation

Answer. True.

Derivation/proof. Each string is uniquely determined by the three one-positions $i < j < k$. Enumerate by maximum position $k = 3, 4, \ldots$; for fixed k there are only $\binom{k-1}{2}$ choices for the other two positions, a finite layer. A countable union of finite layers is countable. The triples $(1, 2, k)$ for $k \geq 3$ provide infinitely many distinct strings, so the set is countably infinite.

S040

Countability

2025 Spring Final | Question 1(a)(i) | Source page 2 | Difficulty 2 | 1 mark
Q-2025S-FINAL-P01-a-i | Answer provenance: official

> (a) [3 pts, 1 each] Determine whether the following statements are True or False. No argument is needed. $\mathcal{P}(X)$ denotes the power set of X.
>
> (i) There exists a countably infinite set X with $\mathcal{P}(X)$ uncountable.

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(a) i) T

Supplementary Explanation

Answer. True.

Derivation/proof. Take $X = \mathbb{N}$, a countably infinite set. If its power set were listed as $A_0, A_1, \ldots$, construct $D = \{n : n \notin A_n\}$. For every j, $j \in D$ exactly when $j \notin A_j$, so $D \neq A_j$. But D is itself a subset of the natural numbers, contradicting completeness of the list. Thus this X has an uncountable power set and witnesses the existential statement.

S041

Countability

2025 Spring Final | Question 1(a)(ii) | Source page 2 | Difficulty 2 | 1 mark
Q-2025S-FINAL-P01-a-ii | Answer provenance: official

> (a) [3 pts, 1 each] Determine whether the following statements are True or False. No argument is needed. $\mathcal{P}(X)$ denotes the power set of X.
>
> (ii) There exists a countably infinite set X with $\mathcal{P}(X)$ also countably infinite.

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(a) ii) F

Supplementary Explanation

Answer. False.

Derivation/proof. Let $X = \{x_0, x_1, \ldots\}$ be any countably infinite set. If its power set had a list $B_0, B_1, \ldots$, form $E = \{x_j : x_j \notin B_j\}$. This subset of X disagrees with each B_j at element x_j, so it is absent from the list. The contradiction applies to every countably infinite X. Hence no set satisfies the proposed existence claim.

S042

Countability

2025 Spring Midterm | Question 4(i) | Source page 6 | Difficulty 2
Q-2025S-MID-P04-i | Answer provenance: official

> Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.
>
> (i) The set of all polynomials with integer coefficients, i.e., polynomials of the form $a_0 + a_1x + \cdots + a_{k-1}x^{k-1} + a_kx^k$ for some $k \in \mathbb{N}$, where $a_i \in \mathbb{Z}$ for $i = 0, \ldots, k$.

Official Answer

Source: physical PDF page 6 (COMP2711_S2025_Midterm.pdf).

(i) Countably infinite. The union of countably many countable sets is countable. This can be proved using the same argument as showing $\mathbb{N}^2$ is countable.

Supplementary Explanation

Answer. Countably infinite.

Derivation/proof. For fixed k, coefficient tuples lie in $\mathbb{Z}^{k+1}$. A finite product of countable sets is countable: enumerate the tuples by the sum of the absolute values of their coordinates. All finite-degree polynomials are in the image of $\bigcup_{k\geq 0} \mathbb{Z}^{k+1}$, a countable union of countable sets. Distinct integer constants give infinitely many different polynomials. The set is therefore countably infinite, not finite.

S043

Countability

2025 Spring Midterm | Question 4(ii) | Source page 6 | Difficulty 2
Q-2025S-MID-P04-ii | Answer provenance: official

> Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.
>
> (ii) The set of all polynomials with integer coefficients and the degree is at most 5.

Official Answer

Source: physical PDF page 6 (COMP2711_S2025_Midterm.pdf).

(ii) Countably infinite.

Supplementary Explanation

Answer. Countably infinite.

Derivation/proof. Write each polynomial uniquely as $a_0 + a_1x + \cdots + a_5x^5$, filling unused coefficients with zero. It corresponds to a tuple in $\mathbb{Z}^6$. Enumerate the tuples by the sum of the absolute values of their coordinates; each layer is finite, hence the collection is countable. Constants $f(x) = j$ for integer j give an infinite subset, so it is countably infinite.

S044

Countability

2025 Spring Midterm | Question 4(iii) | Source page 6 | Difficulty 2
Q-2025S-MID-P04-iii | Answer provenance: official

> Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.
>
> (iii) The set of all polynomials with real coefficients.

Official Answer

Source: physical PDF page 6 (COMP2711_S2025_Midterm.pdf).

(iii) Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Each real t gives the constant polynomial $f_t(x) = t$. Distinct real parameters give distinct polynomials, defining an injection from $\mathbb{R}$ into this set. If the polynomial set were countable, its constant subset would be countable, contradicting the uncountability of $\mathbb{R}$. Allowing additional degrees cannot change this conclusion.

S045

Countability

2025 Spring Midterm | Question 4(iv) | Source page 6 | Difficulty 2
Q-2025S-MID-P04-iv | Answer provenance: official

> Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.
>
> (iv) The set of all polynomials with real coefficients and the degree is at most 5.

Official Answer

Source: physical PDF page 6 (COMP2711_S2025_Midterm.pdf).

(iv) Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. The set still contains every constant polynomial. The map $t \mapsto (t, 0, 0, 0, 0, 0)$ injects the real numbers into its six coefficient positions. If these tuples could be enumerated, projecting onto the first coefficient would enumerate every real number, a contradiction. Thus finitely many real coefficient positions already give an uncountable collection; the degree bound does not make it finite or countably infinite.

S046

Countability

2025 Spring Midterm | Question 4(v) | Source page 6 | Difficulty 2
Q-2025S-MID-P04-v | Answer provenance: official

> Determine whether each of these sets is finite, countably infinite, or uncountable. No explanation is needed.
>
> (v) The set of real roots of $1 - 2x + 3x^2 - 4x^3 + 5x^4$.

Official Answer

Source: physical PDF page 6 (COMP2711_S2025_Midterm.pdf).

(iv) Finite.

Supplementary Explanation

Answer. Finite.

Derivation/proof. Let $f(x) = 1 - 2x + 3x^2 - 4x^3 + 5x^4$. A nonzero quartic has at most four distinct real roots. More strongly,

$$f(x) = x^2(5x^2 - 4x + 1) + (2x^2 - 2x + 1),$$

where $5x^2 - 4x + 1 = 5(x - 2/5)^2 + 1/5 > 0$ and $2x^2 - 2x + 1 = 2(x - 1/2)^2 + 1/2 > 0$. Hence there are no real roots: the set is empty and therefore finite. This calculation is an independent check, not an expansion of the course taxonomy.

S047

Countability

2024 Fall Final | Question 2(b) | Source page 6 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P02-b | Answer provenance: official

> (b) We know that any number of form $\sqrt[n]{p}$ is irrational when p is prime and n is an integer $n \geq 2$. How many irrational numbers of this form exist? Is it finite, countable infinite or uncountable? Briefly argue why. (Note that $y = \sqrt[n]{p}$ means $y^n = p$)

Official Answer

Source: physical PDF page 6 (COMP2711_F2024_Final.pdf).

> (b) Note that there are countably infinite prime numbers and countably infinite values for n. This implies that there are $|\mathbb{N}|^2$ many irrational numbers of this form, which is countable infinite.

Supplementary Explanation

Answer. Countably infinite.

Derivation/proof. Enumerate the primes as $p_1, p_2, \ldots$ and degrees as $2, 3, \ldots$. The index pairs (i, n) can be enumerated along diagonals, so all $\sqrt[n]{p_i}$ form an image of a countable set and are at most countable. Fix $n = 2$: distinct primes give distinct positive square roots, since squaring recovers the prime. There are infinitely many primes, so the image is at least infinite. Together these bounds make it countably infinite. Irrationality is supplied by the question and need not be reproved.

S048

Countability

2024 Fall Midterm | Question 3(a)(i) | Source page 7 | Difficulty 2 | 1 mark
Q-2024F-MID-P03-a-i | Answer provenance: official

> Determine whether each of the following infinite sets is countable or uncountable. No explanation is needed.
>
> (i) The set of all java programs!

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> Answer: (a) (i) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. Every Java program is a finite-length string over a finite alphabet. For each length ℓ there are finitely many strings; enumerating lengths from zero makes all finite strings a countable union of finite sets. Syntactically valid programs are a subset, so are at most countable. Arbitrarily many spaces or repeated statements produce infinitely many distinct valid program texts. Thus the stated infinite program set is countably infinite.

S049

Countability

2024 Fall Midterm | Question 3(a)(ii) | Source page 7 | Difficulty 2 | 1 mark
Q-2024F-MID-P03-a-ii | Answer provenance: official

> Determine whether each of the following infinite sets is countable or uncountable. No explanation is needed.
>
> (ii) The set of coordinates (X, Y) in 2D space such that $X \in \mathbb{Q}$ (X is rational number) and $Y \in \mathbb{Q}'$ (Y is irrational)

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (ii) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Fix the rational coordinate $X = 0$. Every irrational y gives a distinct allowed point $(0, y)$. The irrationals are uncountable: if countable, their union with the countable rationals would make the reals countable, a contradiction. Thus the target set contains an uncountable subset in bijection with the irrationals and is itself uncountable.

S050

Countability

2024 Fall Midterm | Question 3(a)(iii) | Source page 7 | Difficulty 2 | 1 mark
Q-2024F-MID-P03-a-iii | Answer provenance: official

> Determine whether each of the following infinite sets is countable or uncountable. No explanation is needed.
>
> (iii) The set of coordinates (X, Y) in 2D space such that both X, Y are rational numbers.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (iii) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. Enumerate the rationals as $q_1, q_2, \ldots$. Every point is (q_i, q_j). Enumerate by diagonals $i + j = 2, 3, 4, \ldots$; each diagonal contains finitely many index pairs and every positive pair occurs. Thus $\mathbb{Q} \times \mathbb{Q}$ is at most countable. It contains all points $(q_i, 0)$ and is infinite, so it is countably infinite.

S051

Countability

2023 Fall Midterm | Question 3(b)(ii) | Source page 7 | Difficulty 2 | 1 mark
Q-2023F-MID-P03-b-ii | Answer provenance: official

> Without the need for providing any explanation, determine if the following sets are infinitely countable (C) or infinitely uncountable (U):
>
> (ii) The set of all functions from $\{1, 2\} \to \mathbb{N}$

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Midterm.pdf).

> (b) C, C, U

Supplementary Explanation

Answer. C: infinitely countable.

Derivation/proof. The map $f \mapsto (f(1), f(2))$ is a bijection from the function set to $\mathbb{N}^2$: every ordered pair defines exactly one function on the two-element domain. Listing pairs along finite diagonals of increasing coordinate sum enumerates all functions. There are infinitely many choices for $f(1)$ even when $f(2)$ is fixed, so the set is infinite as well as countable.

The second entry of the printed key agrees with the official classification for this atomic item: it is infinitely countable. Thus the set is infinitely countable.

S052

Countability

2023 Spring Midterm | Question 3(1) | Source page 6 | Difficulty 2
Q-2023S-MID-P03-1 | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> 1. $\mathbb{N}^2$

Official Answer

Source: physical PDF page 6 (COMP2711_S2023_Midterm.pdf).

> (a) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. For each possible sum s, list all pairs $(a, b) \in \mathbb{N}^2$ with $a + b = s$. Each diagonal is finite; scanning the diagonals in increasing s reaches every pair because each pair has a finite sum. This constructs an enumeration. There are infinitely many distinct pairs with one coordinate fixed, so the set is countably infinite. The argument works with either convention for whether 0 belongs to $\mathbb{N}$.

The enumeration eventually reaches every pair, so the set is countably infinite. The fixed second coordinate used to show infinitude may be any legal natural number.

S053

Countability

2023 Spring Midterm | Question 3(3) | Source page 6 | Difficulty 2
Q-2023S-MID-P03-3 | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> 3. The set of all functions from $\{0, 1\}$ to $\mathbb{N}$

Official Answer

Source: physical PDF page 6 (COMP2711_S2023_Midterm.pdf).

> (c) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. Associate each function f with the ordered pair $(f(0), f(1))$. This assignment is injective because equal pairs mean equal values at every input, and surjective because any pair of naturals defines such a function. The pairs can be enumerated by increasing sum of their coordinates, so this bijection proves the function set is countably infinite.

The correspondence $f \mapsto (f(0), f(1))$ is a bijection: (a, b) has the inverse function $f(0) = a, f(1) = b$. Therefore the function set is countably infinite.

S054

Countability

2022 Midterm | Question 5(a) | Source page 2 | Difficulty 2
Q-2022-MID-P05-a | Answer provenance: official

> For each of the following sets, determine if it is countable or uncountable. No justification is needed.
>
> $\{x \in \mathbb{R} \mid x^2 \in \mathbb{Z}\}$

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (a) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. A real square is nonnegative, so the set is exactly {0,±√1,±√2,±√3,...}. For each nonnegative integer there are at most two corresponding real roots. This explicitly enumerates the whole set, and the infinitely many positive square roots are distinct. The elements need not be rational for the set to be countable.

Explicitly $x = \pm\sqrt{k}$ for an integer $k \geq 0$, and an enumeration is $0, 1, -1, \sqrt{2}, -\sqrt{2}, \ldots$. The positive roots are distinct, so the set is countably infinite.

S055

Countability

2022 Midterm | Question 5(c) | Source page 2 | Difficulty 2
Q-2022-MID-P05-c | Answer provenance: official

> For each of the following sets, determine if it is countable or uncountable. No justification is needed.
>
> The union of countably many countable sets.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (c) Countable

Supplementary Explanation

Answer. Countable.

Derivation/proof. List the sets A0,A1,... and choose an enumeration a_(i,0),a_(i,1),... for each nonempty one, repeating values for finite sets if needed. Enumerate index pairs (i,j) by increasing i+j and output the corresponding element, skipping repetitions. Every member occurs at a finite pair and hence a finite stage, so the union is at most countable; it may be finite or empty.

Visit index pairs in increasing order of $i + j$, with finitely many pairs at each stage. Every union member is output at a finite stage. The union may also be finite or empty.

S056

Countability

2020 Midterm | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-MID-P05-b | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> $P(\mathbb{N})$

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Suppose all subsets can be listed $S_0, S_1, \ldots$ using an enumeration of the naturals. Define $D = \{i : i \notin S_i\}$. For every i, membership of i in D differs from membership in S_i, so $D \neq S_i$. Yet D is a subset of the naturals, contradicting completeness of the list.

Index the enumeration from whichever natural-number starting convention is adopted; the diagonal contradiction is unchanged.

More explicitly, set $D = \{i \in \mathbb{N} : i \notin S_i\}$. If the purported exhaustive list had $D = S_j$, then $j \in D \iff j \notin S_j$ would contradict that equality. Thus the diagonal subset is missing from every countable list.

S057

Countability

2020 Midterm | Question 5(c) | Source page 3 | Difficulty 2
Q-2020-MID-P05-c | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> $P(P(\mathbb{N}))$

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. The map $S \mapsto \{S\}$ injects $P(\mathbb{N})$ into its power set. Cantor's diagonal argument shows $P(\mathbb{N})$ is uncountable. If $P(P(\mathbb{N}))$ were countable, every subset, including that singleton image, would be countable; its bijection with $P(\mathbb{N})$ would give a contradiction.

If $\{S\} = \{T\}$, equality of the unique elements gives $S = T$. Hence the singleton map is injective, and its injective image is an uncountable subset of the target power set.

S058

Countability

2019 Midterm | Question 7(c) | Source page 9 | Difficulty 2 | 3 marks
Q-2019-MID-P07-c | Answer provenance: official

> Give an example of two uncountable sets A and B such that $A - B$ is
>
> (c) uncountable.

Official Answer

Source: physical PDF page 9 (2019-midterm.pdf).

In each case, let us take A to be the set of real numbers. We can let B be the set of positive real numbers. Then $A - B$ is the set of negative real numbers and 0, which is certainly uncountable.

Supplementary Explanation

Answer. $A = \mathbb{R}$, $B = (0, \infty)$; then $A - B = (-\infty, 0]$.

Derivation/proof. The positive real interval B is uncountable: for instance $(0, 1) \subseteq B$ is uncountable. The whole real line A is uncountable. The difference contains the negative interval $(-1, 0)$, which is in bijection with $(0, 1)$ by negation and hence uncountable. Thus the difference has the required cardinality.

The intervals $(0, 1) \subseteq B$ and $(-1, 0) \subseteq A - B$ independently establish the two required uncountability claims.

S059

Countability

2024 Fall Midterm | Question 3(a)(iv) | Source page 7 | Difficulty 3 | 1 mark
Q-2024F-MID-P03-a-iv | Answer provenance: official

> Determine whether each of the following infinite sets is countable or uncountable. No explanation is needed.
>
> (iv) The set $\mathcal{P}(\mathbb{R}) - \mathbb{R}$, where $\mathbb{R}$ is the set of real numbers and $\mathcal{P}(\mathbb{R})$ is its power set.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (iii) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Suppose for contradiction that $C = \mathcal{P}(\mathbb{R}) \setminus \mathbb{R}$ is countable. The remaining part $D = \mathcal{P}(\mathbb{R}) \cap \mathbb{R}$ is a subset of the reals, so has cardinality at most $|\mathbb{R}|$. Hence $\mathcal{P}(\mathbb{R}) = C \cup D$ has cardinality at most $|\mathbb{R}|$, since adding countably many elements to an infinite real set does not increase its cardinality. Cantor's theorem gives $|\mathcal{P}(\mathbb{R})| > |\mathbb{R}|$, a contradiction. Thus the difference is uncountable, without treating all reals as all real subsets.

S060

Countability

2023 Fall Midterm | Question 3(b)(iii) | Source page 7 | Difficulty 3 | 1 mark
Q-2023F-MID-P03-b-iii | Answer provenance: official

> Without the need for providing any explanation, determine if the following sets are infinitely countable (C) or infinitely uncountable (U):
>
> (iii) The set of all functions from $\mathbb{N} \to \{1, 2\}$

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Midterm.pdf).

> (b) C, C, U

Supplementary Explanation

Answer. U: infinitely uncountable.

Derivation/proof. Suppose these functions had a complete list $f_1, f_2, \ldots$. Define $g(n) = 3 - f_n(n)$, with a harmless reindexing if naturals start at 0. Because $f_n(n)$ is either 1 or 2, g also maps into $\{1, 2\}$. However, $g(n) \neq f_n(n)$ for every n, so g differs from each listed function. This contradicts completeness of the list and proves uncountability.

This agrees with the official classification, the third entry U. It contradicts a complete enumeration.

S061

Countability

2023 Fall Midterm | Question 3(c) | Source page 7 | Difficulty 3 | 3 marks
Q-2023F-MID-P03-c | Answer provenance: official

> An algebraic number is a number that is a root of a non-zero polynomial in one variable with integer coefficients. As an example, $(1+\sqrt{5})/2$ is an algebraic number, because it is a root of the polynomial $x^2 - x - 1$. Or, the complex number $i+1$ is an algebraic number as it is a root of $x^4 + 4$. Briefly argue why we have countably many algebraic numbers.
>
> Hint: Try to think about how many polynomials exist and how many roots each one has. You may use the result of any of (b) as well.

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Midterm.pdf).

> (c) You can express the set of algebraic numbers as a countable union of finite sets.

Supplementary Explanation

Answer. The algebraic numbers form a countably infinite set.

Derivation/proof. For every degree d, a polynomial is specified by a tuple $(a_0, \ldots, a_d) \in \mathbb{Z}^{d+1}$ with $a_d \neq 0$, and finite integer tuples form a countable set. Taking the union over nonnegative d leaves countably many nonzero polynomials. Each degree-d polynomial has at most d distinct complex roots. Thus the algebraic numbers are a countable union of finite root sets, hence at most countable. Every integer k is a root of $x - k$, so the set is infinite and therefore countably infinite.

The nonzero condition is essential: the zero polynomial would make every complex number a root.

A degree-d nonzero polynomial has at most d distinct roots.

S062

Countability

2023 Spring Midterm | Question 3(2) | Source page 6 | Difficulty 3
Q-2023S-MID-P03-2 | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> 2. $P(\mathbb{N})$

Official Answer

Source: physical PDF page 6 (COMP2711_S2023_Midterm.pdf).

> (b) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Suppose all subsets were listed as $A_1, A_2, \ldots$ (reindex naturals starting at 1 if necessary). Define $D = \{n : n \notin A_n\}$. For every index k, membership of k in D is opposite to its membership in A_k, so $D \neq A_k$. Yet D is a subset of $\mathbb{N}$ and must occur in a complete list. This contradiction proves uncountability.

The sets differ at least on element k, contradicting a complete list of all subsets.

S063

Countability

2023 Spring Midterm | Question 3(4) | Source page 6 | Difficulty 3
Q-2023S-MID-P03-4 | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> 4. The set of all functions from $\mathbb{N}$ to $\{0, 1\}$

Official Answer

Source: physical PDF page 6 (COMP2711_S2023_Midterm.pdf).

> (d) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Map f to $A_f = \{n : f(n) = 1\}$. Every subset of naturals determines exactly one such indicator function, and distinct subsets produce distinct functions. Thus this is a bijection with $P(\mathbb{N})$. Alternatively, from any list $f_1, f_2, \ldots$ define $g(n) = 1 - f_n(n)$; then g differs from every listed function. Hence the set is uncountable.

The new function contradicts a complete enumeration.

S064

Countability

2023 Spring Midterm | Question 3(5) | Source page 6 | Difficulty 3
Q-2023S-MID-P03-5 | Answer provenance: official

> Decide if the following sets are countable or uncountable. Let $\mathbb{N}$ be the set of natural numbers. $P(S)$ denotes the power set of S. No need to justify your answers.
>
> 5. The set of all functions from $\mathbb{N}$ to $\mathbb{N}$

Official Answer

Source: physical PDF page 6 (COMP2711_S2023_Midterm.pdf).

> (e) Uncountable.

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. Choose two distinct natural numbers, for example 1 and 2. All functions from $\mathbb{N}$ to $\{1, 2\}$ are also functions from $\mathbb{N}$ to $\mathbb{N}$. The two-valued function set is uncountable by diagonalization: reverse the nth output of the nth listed function. If the larger function set were countable, every subset would be at most countable, contradicting this embedded uncountable subset.

For the subfamily with range $\{1, 2\}$, define $g(n) = 3 - f_n(n)$; it still has an allowed value and differs from each listed function. If the larger family were countable, every subset would be at most countable, a contradiction.

S065

Countability

2022 Midterm | Question 5(b) | Source page 2 | Difficulty 3
Q-2022-MID-P05-b | Answer provenance: official

> For each of the following sets, determine if it is countable or uncountable. No justification is needed.
>
> $\{x \in \mathbb{R} \mid \text{there is a decimal expansion of } x \text{ with only even digits}\}$

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (b) Uncountable

Supplementary Explanation

Answer. Uncountable.

Derivation/proof. For every infinite binary sequence b1,b2,... map it to decimal 0.(2b1)(2b2).... All its digits are 0 or2 and hence even. If two sequences first differ at place k, the leading difference is 2·10^(−k), while the maximum possible tail difference is 2∑_(j>k)10^(−j)=(2/9)10^(−k), smaller. Thus the map is injective. Binary sequences are uncountable by diagonalization, so this subset and hence the source set are uncountable.

The injection is $x = \sum_{i \geq 1} 2b_i 10^{-i}$. A first difference $2 \cdot 10^{-k}$ exceeds the largest tail difference $\frac{2}{9}10^{-k}$, so the resulting real numbers are still different.

S066

Countability

2021 Midterm | Question 4(a) | Source page 2 | Difficulty 3 | 6 marks
Q-2021-MID-P04-a | Answer provenance: official

> Let $\mathbb{N}$ be the set of all natural numbers.
>
> Show that $\{S \mid S \subseteq \mathbb{N}, |S| \text{ is finite}\}$ is countable.

Official Answer

Source: physical PDF page 2 (2021-midterm.pdf).

> (a) Here is one enumeration method for $\{S \mid S \subseteq \mathbb{N}, |S| \text{ is finite}\}$. We first enumerate $\varnothing$. Then for $k = 0, 1, \ldots$, we enumerate all $S \subseteq \mathbb{N}$ such that $\max S = k$. For each k, there are finitely many such S, and we can enumerate them in any order.

Supplementary Explanation

Answer. The family is countably infinite.

Derivation/proof. List the empty set first. For each natural k in increasing order, list every finite subset whose maximum is k. Each such block is finite because its elements come from the finite initial segment through k; if naturals begin at zero, the block has 2^k members. Every nonempty finite set has a finite maximum and appears in exactly one block, giving an enumeration. Singleton sets show the family is infinite.

Each group occurs after finitely many earlier entries. For naturals starting at zero, k must be included and each of the other k elements may be chosen freely, giving 2^k sets in the group.

To enumerate the family, list the empty set first. Then every nonempty finite subset has a maximum, so the enumeration covers it. The family is countably infinite.

S067

Countability

2019 Midterm | Question 7(b) | Source page 9 | Difficulty 3 | 3 marks
Q-2019-MID-P07-b | Answer provenance: official

> Give an example of two uncountable sets A and B such that $A - B$ is
>
> (b) countably infinite.

Official Answer

Source: physical PDF page 9 (2019-midterm.pdf).

In each case, let us take A to be the set of real numbers. We can let B be the set of real numbers that are not positive integers; in symbols, $B = A - \mathbb{Z}^+$. Then, $A - B = \mathbb{Z}^+$, which is countably infinite.

Supplementary Explanation

Answer. $A = \mathbb{R}$, $B = \mathbb{R} \setminus \mathbb{Z}_{>0}$; then $A - B = \mathbb{Z}_{>0}$.

Derivation/proof. The difference is exactly the positive integers, so it is countably infinite. The input A is uncountable. If B were countable, $\mathbb{R} = B \cup \mathbb{Z}_{>0}$ would be a union of two countable sets and therefore countable, a contradiction. Thus B is also uncountable, verifying every requirement.

For these sets, $A - B = \mathbb{Z}^+$, which is countable. If B were countable too, $\mathbb{R} = B \cup \mathbb{Z}^+$ would be countable, contradicting the uncountability of the reals.

S068

Countability

2019 Midterm | Question Bonus Problem | Source page 15 | Difficulty 4 | 10 marks
Q-2019-MID-BONUS | Answer provenance: official

> An enemy tank is moving at a constant speed along a road. You can imagine the road as the infinite number line, where every position is identified by a number. The tank starts at time 0 at an unknown location x and moves at an unknown speed of y meters a minute, but we know that x and y are both positive integers. We can fire a cannon every minute at a specific location, but cannot see the tank. Can you design a strategy that guarantees to hit the tank eventually?

Official Answer

Source: physical PDF page 15 (2019-midterm.pdf).

The Cartesian product $\mathbb{Z}^+ \times \mathbb{Z}^+$ is countable, and we can enumerate the pairs $(x, y) \in \mathbb{Z}^+ \times \mathbb{Z}^+$ one by one. At time i, we use the i-th pair (x, y) in this enumeration to decide the firing location. Note that if (x, y) are correct, the tank will be at position $x + y \cdot i$, so we fire at this location. Because every pair (x, y) will eventually be enumerated, we are guaranteed to hit the tank.

Supplementary Explanation

Answer. Enumerate all positive integer pairs (x_i, y_i); at minute i, fire at $x_i + iy_i$.

Derivation/proof. Enumerate pairs by increasing sum $x + y$, and within each finite diagonal by increasing x: $(1, 1), (1, 2), (2, 1), \ldots$. Every positive integer pair occurs at a finite index because only finitely many pairs have smaller sum. If the true parameters occur at index i, the tank's position at that minute is exactly $x_i + iy_i$, the chosen firing point. Hence a hit occurs no later than that finite minute. This is a deterministic exhaustive strategy, not a claim of a uniform time bound over all parameters.

Use the diagonals $x + y = 2, 3, 4, \ldots$, each of which contains finitely many positive pairs. When the true pair is enumerated as (x_j, y_j), the tank at time j is at $x + jy = x_j + jy_j$, precisely the chosen firing location.

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